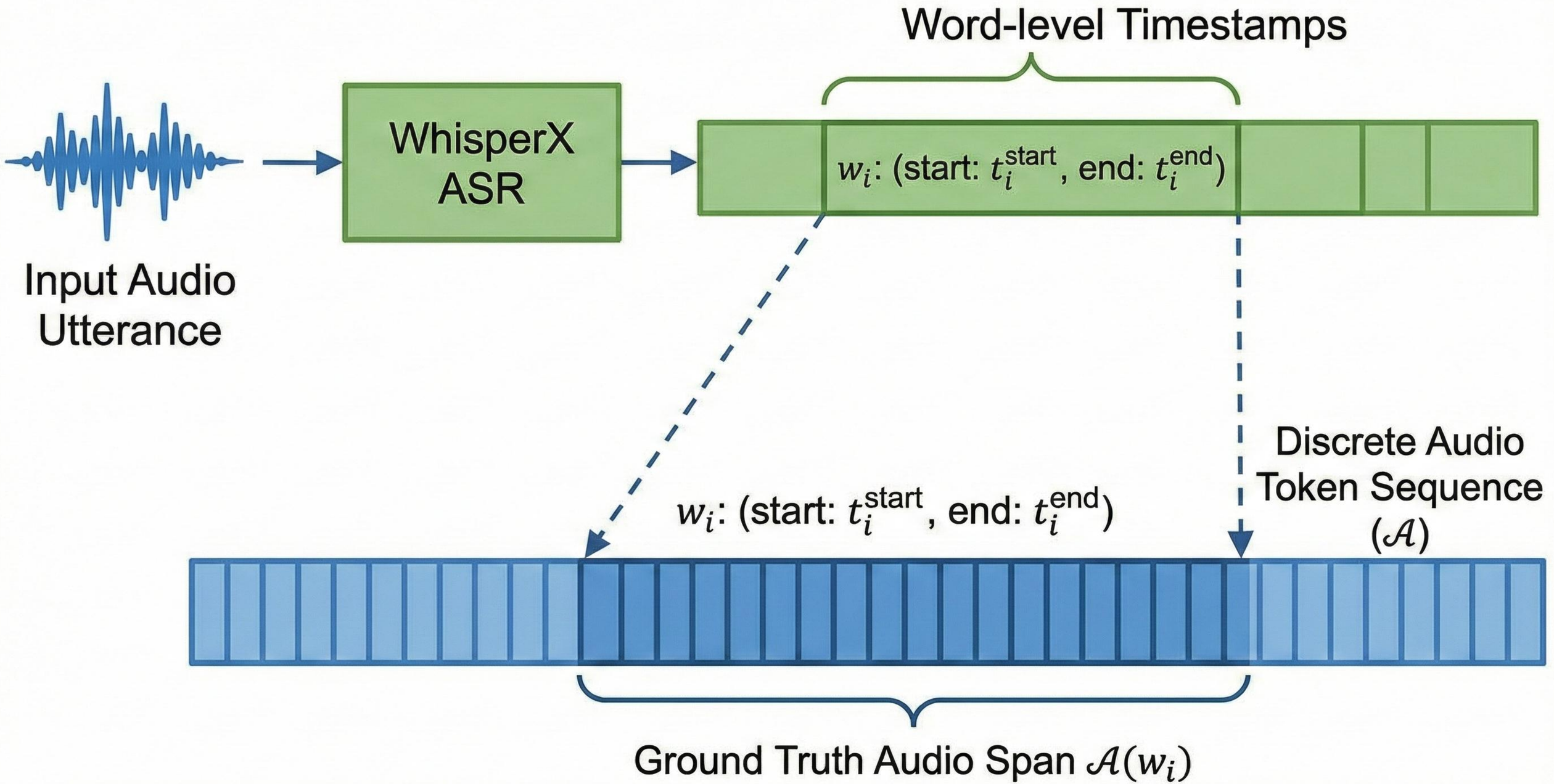

Word-level Timestamps
WhisperX ASR
Input Audio Utterance
w_i: (start: t_i^{start}, end: t_i^{end})
w_i: (start: t_i^{start}, end: t_i^{end})
Discrete Audio Token Sequence ($\mathcal{A}$)
Ground Truth Audio Span $\mathcal{A}(w_i)$